

Springer Texts in Statistics

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An Introduction to Statistical Learning

with Applications in R

 Springer

ANALYTICS

Week 2/3

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*Examples Inspired By
CAS, SOA, Coaching
Actuaries ©*

Key Formulas This Week

- Basic Linear Regression: $b_1 = r \frac{s_y}{s_x} = \frac{Cov(X,Y)}{s_x^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$, $b_0 = \bar{y} - b_1 \bar{x}$
- $\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$
- $SST = SSE + SSReg$
- Residual: $e_i = y_i - \hat{y}_i$, $RSS = SSE = \sum e_i^2$
- $MSE = s^2 = \frac{1}{n-2} \sum_{i=1}^n e_i^2 = \frac{SSE}{n-2}$

Key Formulas This Week 2

- $s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} (\sum_{i=1}^n x_i^2 - n\bar{x}^2)$
- $s_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}$
- $r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$
- Coefficient of determination: $R^2 = \frac{SSReg}{SST}$

Model Building Process

- 1. Clean, analyze, transform data.
- 2. Hypothesize model structures.
- 3. Estimate parameter coefficients on training.
- 4. Test parameters for statistical significance.
- 5. Create alternate models.
- 6. Use model selection to pick best candidates.
- 7. Check assumptions w/model diagnostics.
- 8. Vet best candidates on test data.
- 9. Make predictions, create intervals based off the chosen final model.

Basic Linear Regression

- In basic linear regression, the true model is assumed to be:
- $Y = \beta_0 + \beta_1 x + \varepsilon$
- The fitted regression model is:
- $\hat{y} = b_0 + b_1 x$
 - β_0, b_0 : Intercept
 - β_1, b_1 : Slope
 - ε : Error / Disturbance term
- The line will always pass through _____

Basic Linear Regression 2

- $\beta_0, \beta_1, \sigma^2$ refer to the true (unknown) population parameters
- b_0, b_1, s^2 refer to the sample estimated statistics used to infer
- The estimated parameters are called **least-squares estimators** and are equivalent to the **maximum likelihood estimators** for basic linear regression and are **unbiased**.
- Unbiased: $E[b_1] = \beta_1, E[b_0] = \beta_0$

Linear Model Assumptions

- Basic linear regression model assumptions:
 - 1. $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$
 - 2. x_i 's are **non-stochastic variables**.
 - 3. $E[\varepsilon_i] = 0$
 - 4. $Var(y_i) = \sigma^2$ or equivalently $Var(\varepsilon_i) = \sigma^2$
- This property of constant variance of y_i is known as

Linear Model Assumptions 2

- Basic linear regression model assumptions:
- 5. y_i' s (or equivalently ε_i' s) are _____
- 6. y_i' s (or equivalently ε_i' s) are _____

Statistics Refresher

- Recall from probability and statistics:
- Correlation: $r = \frac{Cov(X,Y)}{s_x s_y}$
- $Cov(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$
- Sample variance of x : $s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
- Sample variance of y : $s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$

BLR Parameter Estimation

- Coefficients are estimated using ordinary least squares, by minimizing $\sum_{i=1}^n (y_i - \hat{y}_i)^2$.
- $\hat{y} = b_0 + b_1 x$
- Slope estimate: $b_1 =$

- Intercept estimate: $b_0 = \bar{y} - b_1 \bar{x}$

Algebraic Equivalencies

- These may help you with algebraic manipulation Qs.
- $\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$
- $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}$

Calculator Tutorials

- [TI-30XS](#)
- [TI-30X IIS](#)
- [BA II Plus](#)

Parameter Interpretation

- $\hat{y} = b_0 + b_1x$
- b_0 estimates $\hat{y}$ when _____
- b_1 is the change in $\hat{y}$ per unit change in x and the _____

Matrix Notation

- $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ in matrix notation is:

- $$\begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \dots & \dots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_n \end{bmatrix}$$

- $$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Matrix Notation 2

- To solve for the parameters and make predictions,
- $\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$
 - $\mathbf{X}^T$: transpose of the $\mathbf{X}$ matrix, meaning rows become columns and vice-versa
 - $\mathbf{A}^{-1}$: inverse of matrix $\mathbf{A}$
- $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$, where $\mathbf{H}$ is the **hat** matrix
- $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$

Poll Concept Check #1

- Which of the following statements are true regarding a simple linear regression model?
- I. The model equation is: $Y = \beta_0 + \beta_1 x$
- II. The mean of the response is modeled to vary based on the observations.
- III. The variance of the response is modeled to vary based on the observations.
- IV. Estimated coefficients b_0, b_1 are unbiased maximum likelihood estimators.

1. Calculator Illustration

- For the following dataset, use the method of least squares to estimate b_0, b_1 for $Y = \beta_0 + \beta_1 X + \varepsilon$.

i	1	2	3
x_i	2	-6	7
y_i	3	4	6

1. Calculator Illustration 2

- If given data, never do these calculations by hand. Use 2-Var Stats on your calculator.
- “data” button. Input x in L1, y in L2
- “stat” button (2^{nd} + data). Then, 2-Var Stats

1. Calculator Illustration 3

- 1: $n = 3$
- 2: $\bar{x} = 1$
- 3: $s_x = 6.5574$
- 5: $\bar{y} = 4.\bar{3}$
- 6: $s_y = 1.5275$
- D: $b_1 = a = 0.1279$
- E: $b_0 = b = 4.2054$
- F: $r = 0.5491$

Example 1

- You are fitting the linear regression model:
 $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ to 10 points of data to model pounds of food eaten using hours spent hiking as the predictor.
- $\sum x_i = 110, \sum y_i = 150$
- $\sum x_i^2 = 1900, \sum y_i^2 = 5000, \sum x_i y_i = 2000$
- $b_1 = 0.5072$
- A. Estimate the pounds of food eaten on 9 hours of hiking.
- B. Determine the first integer hour that the estimated food eaten exceeds 12 pounds.
- C. Verify b_1 using the formula.

Example 1 Sol

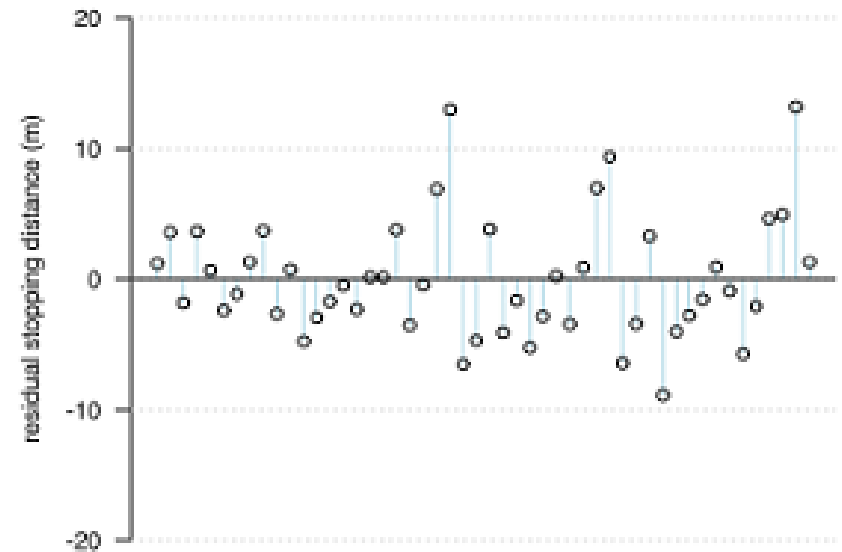
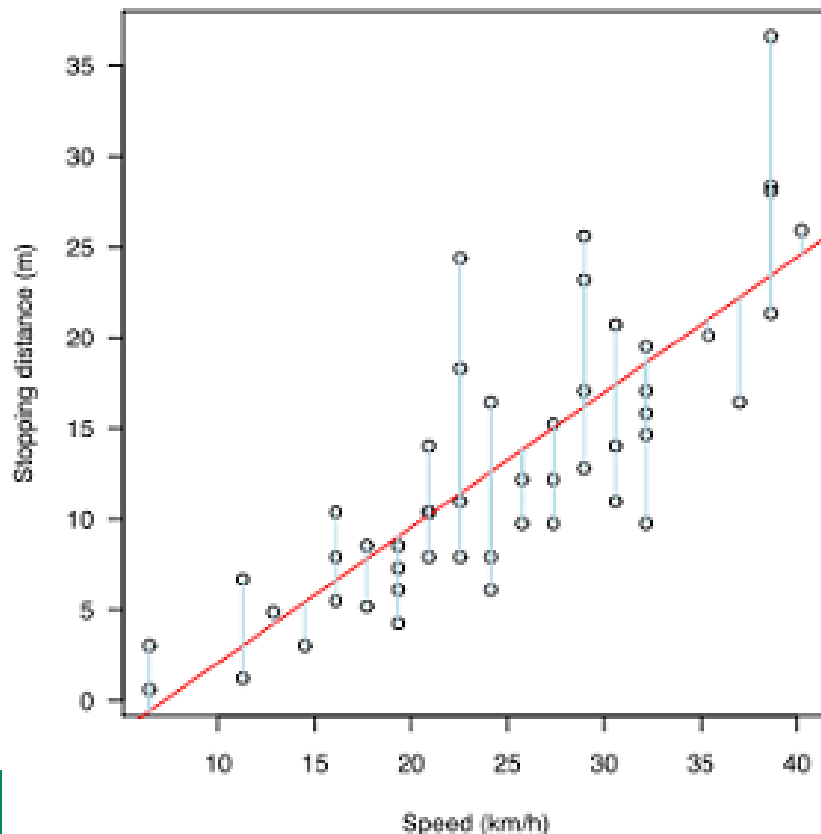
- $b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$
- $b_0 = \bar{y} - b_1 \bar{x}$
- $\hat{y} = b_0 + b_1 x_{pred}$

Residuals

- Residual: $e_i =$
- $E[e_i] = 0$ and $\sum_i x_i e_i = 0$
- Positive Residual $\rightarrow$
- Negative Residual $\rightarrow$ Actual $<$ Predicted
- True error terms ε_i are unobservable, but residuals e_i are observable after fitting.

Residuals Plot

- <https://study.com/academy/lesson/residual-plots-definition-example.html>



Residuals 2

- Analyzing residuals can help us improve the model, as well as _____
- Another name for the error sum of squares is the residual sum of squares: $RSS = SSE = \sum e_i^2$

MSE

- Mean squared error (**MSE**) is the estimate of σ^2 and in basic linear regression,
- $MSE = s^2 = \frac{\sum_{i=1}^n e_i^2}{n-2} =$
- Degrees of freedom: $df = n - 2$
- Residual standard error: $s = RSE = \sqrt{MSE} = \sqrt{\frac{RSS}{n-2}}$

SST

- Total sum of squares represents the variation in the response y'_i s.
- $SST =$
- $s_y^2 = \frac{SST}{n-1}$ is the _____

SSE

- Error sum of squares (SSE) represents the variation in the y'_i s after _____
- $SSE =$

Sum of Squares Decomp. *

- $y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})$
- It can be algebraically proven that for linear regression,
- $\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$
- $SST =$
 - SST : total variation in y
 - SSE : total variation remaining after use of x
 - $SSReg$ or SSR : variation explained through x

Algebraic Equivalencies 2

- Under basic linear regression only,
- $SSReg = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = b_1^2 s_x^2 (n - 1) = b_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 = b_1 \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$
- Recall: $s_x^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$

Differentiation

- $MSE = s^2 = \frac{SSE}{n-2} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}$ is the mean squared error for the linear regression model
- $s_y^2 = \frac{SST}{n-1} = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}$ is the sample variance of the y_i
- $s_x^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$ is the sample variance of the x_i

Correlation Coefficient

- The ordinary (Pearson) **correlation coefficient** summarizes the strength of the linear relationship between two variables

- $$r = \frac{Cov(X,Y)}{s_x s_y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

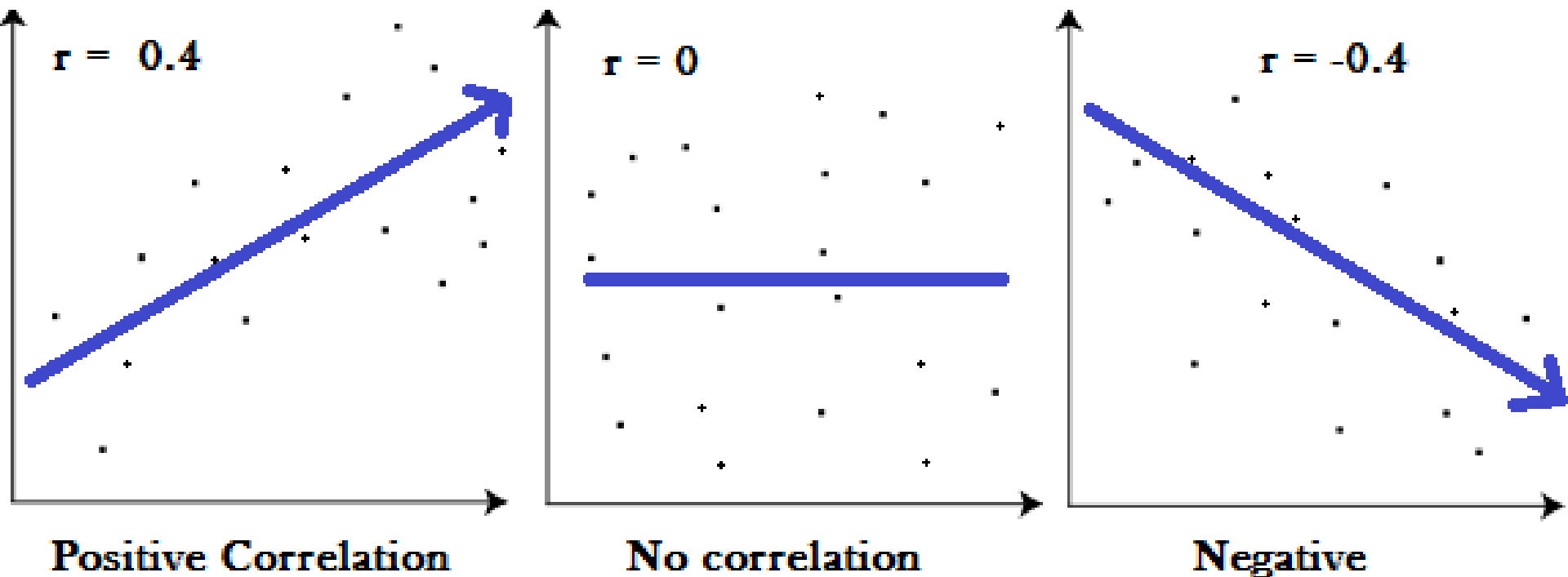
- (x_i, y_i) coordinates for each i
- $\bar{x}, \bar{y}$: averages of x 's, y 's respectively
- n : total # of pairs

Correlation Coefficient 2

- $-1 \leq r \leq 1$
- $r > 0$ is _____
- $r < 0$ is negative correlation
- Larger $|r|$ implies stronger _____
- r is **location & scale invariant**, dimensionless

Correlation Coefficient 3

- <http://www.statisticshowto.com/probability-and-statistics/correlation-coefficient-formula/>



Coefficient of Determination

- Coefficient of determination: $r^2 = R^2 =$
- Represents the proportion of variability in Y explained by the linear relation of y with x
- $0 \leq R^2 \leq 1$
- r^2, R^2 equivalent in basic linear regression only

Poll Concept Check #2

- In a simple linear regression model, which statements are true?
- I. R^2 is the fraction of the variation in y about $\bar{y}$ that is explained by the linear relationship of y with x
- II. A positive residual indicates that the corresponding prediction is under-valued.
- III. A residual is the predicted response minus the actual response for a certain observation.
- IV. The choice of explanatory variable impacts total sum of squares.

2. R^2 Illustration

- A. If $\bar{x} = 10$, $\bar{y} = 20$, $s_x^2 = 100$, $s_y^2 = 400$, $cov_{xy} = -50$, calculate R^2 .
- B. You run a simple linear regression on 5 observations. The residuals of the first 4 observations are $\{-1, 3, 0, -4\}$. The unbiased sample standard deviation of the response is 6. Calculate the coefficient of determination.

2. R^2 Illustration 2

- $\sum_i e_i = 0$
- $SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- $s_y^2 = \frac{SST}{n-1} = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n-1}$
- $R^2 = \left[\frac{Cov(X,Y)}{s_x s_y} \right]^2 = \frac{SSReg}{SST} = 1 - \frac{SSE}{SST}$

2. R^2 Illustration 3

HW Note

- While our class focuses on the quantitative Qs, SRM and the CA HW has a significant # of conceptual Qs.
- The concept Qs you can hone via reading the source text, doing the HWs, and coming to office hours with any lingering questions.

Example 2 (SOA)

- You wish to determine the nature of the relationship between bake sales (y) and the # of flyers posted nearby (x). Data collected on eight consecutive days is shown below.
- Using the method of least squares, you determine:
 - $R^2 = 0.7$
 - $\sum_{i=1}^8 y_i = 50$
 - $\sum_{i=1}^8 y_i^2 = 900$
 - Determine the model mean-squared error.

Example 2 Sol

- $R^2 = \frac{SSReg}{SST}$
- $SST = SSE + SSReg$
- $SST = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - \frac{(\sum_{i=1}^n y_i)^2}{n}$

Summary Check

- *Can you describe in your own words:*
- Key Terms: Mean squared error, residual standard error, total sum of squares, regression of squares, error sum of squares, correlation coefficient, coefficient of determination
- Formulas for b_0 , b_1 , R^2
- Assumptions of simple linear regression